

Simulation Study: Base Model and Sensitivity Scenarios

A Bayesian Framework for Causal Inference with Error-Prone Environmental Exposures

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1 Motivation

The simulation reflects the two-level structure of the application: area-level count health outcomes for participants nested within Forward Sortation Areas (FSA or the first three characters of a Canadian postal code), with area-level exposure assigned from a spatiotemporal prediction model.

2 Notation

- $i = 1, \dots, I$: area index (FSAs)
- A_i : latent true area-level $\text{PM}_{2.5}$ exposure, unobserved
- Z_i : observed error-prone surrogate from a spatiotemporal prediction model (e.g. CanOSSEM)
- $\mathbf{X}_i = (X_{i1}, \dots, X_{i4})^\top$: area-level confounders affecting both exposure and outcome
- D_i : area-level calibration covariate explaining prediction error magnitude in Z_i (e.g. distance to nearest monitoring station, area deprivation)
- Y_i : area-level count health outcome (number of incident depression cases)
- N_i : area population size, used as Poisson offset
- ϕ_i : spatial random effect on the latent exposure surface, following the CAR prior in equation (1)
- $\mathbf{W}$: $I \times I$ binary adjacency matrix ($W_{ij} = 1$ if areas i and j share a border)
- $\mathbf{D}_W = \text{diag}(\mathbf{W}\mathbf{1})$: diagonal degree matrix (number of neighbours per area)
- $\rho_A \in [0, 1)$: spatial correlation in the exposure CAR prior
- σ_ϕ^2 : marginal variance of the spatial random effect ϕ_i

As in the thesis proposal, $\mathbf{X}_i$ controls confounding while D_i indexes the measurement error mechanism. D_i may overlap with $\mathbf{X}_i$ but is conceptually distinct.

3 Key Assumptions

The simulation satisfies the following identification assumptions from the thesis proposal.

- **Non-differential surrogate:** $Y_i \perp Z_i \mid A_i, \mathbf{X}_i, D_i$. Once we condition on the true exposure and covariates, the surrogate carries no additional information about the outcome. Satisfied by construction since Y_i is generated from A_i only and Z_i never appears in the outcome equation.
- **Transportability:** The error relationship is assumed transportable to the target population. When direct calibration data are unavailable, inference about A_i is more model-dependent; this is examined in sensitivity scenario S2.

4 Base Model

4.1 Spatial Structure and CAR Prior

Areas are arranged on a $\sqrt{I} \times \sqrt{I}$ synthetic grid with rook contiguity (shared edges only). $I = 200$ balances the sample sizes used by Josey et al. (2023) ($n \in \{400, 800\}$) against the computational cost of BART.

Following thesis proposal Section 4.3, spatial dependence enters *only* through the latent exposure model via ϕ_i . The spatial random effect follows the Conditional Autoregressive (CAR) prior of Besag (1974):

$$\phi \mid \rho_A, \sigma_\phi^2 \sim \mathcal{N}(\mathbf{0}, \sigma_\phi^2(\mathbf{D}_W - \rho_A \mathbf{W})^{-1}). \quad (1)$$

- ϕ_i is a spatially structured random effect: neighbouring areas have similar ϕ_i values, inducing correlation in their true exposures A_i
- The precision matrix $(\mathbf{D}_W - \rho_A \mathbf{W})$ is sparse with adjacent pairs having off-diagonal entry $-\rho_A$ and non-adjacent pairs having zero
- $\rho_A = 0$: areas are spatially independent; $\rho_A \rightarrow 1$: strong smoothing across neighbours
- $\sigma_\phi^2 = 0.3$ is a design choice giving moderate spatial variation in the exposure surface; in the application this parameter would be estimated from data
- Spatial correlation in the outcome is induced indirectly through A_i ; no separate outcome random effect is added

4.2 Covariates and Population Sizes

- **Area confounders:** $X_{ik} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$ for $k = 1, 2, 3, 4$
- **Population sizes:** $\log(N_i) \sim \mathcal{N}(8.5, 1.2^2)$, giving median $\approx 4,900$, 5th percentile ≈ 600 , 95th percentile $\approx 40,000$. Motivated by Statistics Canada 2021 Census: Ontario has ≈ 500 FSAs with total population 14,223,942, ranging from dense urban FSAs in Toronto to sparse rural FSAs in Northern Ontario.

4.3 Calibration Covariate D_i

D_i captures area characteristics that determine prediction error in Z_i .

Base model: scalar D_i . $D_i \stackrel{\text{iid}}{\sim} \text{Gamma}(2, 1)$ (mean 2, variance 2), representing distance to the nearest monitoring station.

- **Why Gamma?** Distance is strictly positive and right-skewed. Most Ontario FSAs are moderately close to one of the 34 NAPS stations, but some rural areas are very far. Shape 2 gives a unimodal distribution; Exponential (shape 1) would over-represent very small values.
- **Scientific motivation:** Gryparis et al. (2009) and Szpiro et al. (2011) show that kriging prediction error variance is primarily determined by distance to the monitor network. MELONS (Evangelopoulos et al., 2026) identified distance, deprivation, and housing type as empirical determinants of $\text{PM}_{2.5}$ measurement error in London, UK; we use their qualitative finding but not their specific values since they are not directly transferable to Ontario.
- **Ideally:** D_i would be the actual Euclidean distance from each Ontario FSA centroid to the 34 NAPS station coordinates, replacing the Gamma assumption with an empirical distribution.

Extension: vector D_i . In a full implementation D_i would be a vector with components such as: $D_{i1} \sim \text{Gamma}(2, 1)$ (distance to monitor), $D_{i2} \sim \mathcal{N}(0, 1)$ (income and employment indicators as a standardised score), $D_{i3} \sim \text{Beta}(2, 2)$ (proportion detached housing). The error model would be $\omega^2(\mathbf{D}_i) = \exp(\eta_0 + \eta_1 D_{i1} + \eta_2 D_{i3})$. Left as a future extension.

4.4 Latent True Exposure

The latent true exposure is generated from:

$$A_i \mid \mathbf{X}_i, \phi_i \sim \mathcal{N}(7 + 1.5X_{i1} - 1.5X_{i2} - 1.5X_{i3} + 1.5X_{i4} + \phi_i, 9), \quad (2)$$

with $\sigma_A = 3$, where ϕ_i is the spatial random effect following the CAR prior defined in equation (1) with parameters (ρ_A, σ_ϕ^2) . The spatial random effect ϕ_i induces correlation in A_i across neighbouring areas: areas with ρ_A close to 1 have exposures that vary smoothly across the grid, reflecting a regional wildfire plume, while $\rho_A = 0$ gives spatially independent exposures.

- **Mean = 7 $\mu\text{g}/\text{m}^3$:** Empirical annual average $\text{PM}_{2.5}$ in Ontario (mean ≈ 7 , median $\approx 5.4 \mu\text{g}/\text{m}^3$)
- $\sigma_A = 3$: Reflects realistic spatial variation
- **Covariate coefficients ± 1.5 :** Provides meaningful confounding given the exposure distribution
- **Note on skewness:** Ontario $\text{PM}_{2.5}$ is heavily right-skewed (skewness ≈ 9.2 , 99th percentile ≈ 26.4 , extreme values $> 200 \mu\text{g}/\text{m}^3$ during severe wildfire events). The Normal captures typical days but not extreme events.

4.5 Measurement Error Model

Two parameterisations of the measurement error model are considered. Both generate Z_i as a function of the true exposure A_i and calibration covariate D_i , and both satisfy the non-differential assumption by construction since Z_i does not appear in the outcome equation.

In both versions the measurement errors $\varepsilon_i = Z_i - \mathbb{E}[Z_i \mid A_i, D_i]$ are assumed conditionally independent across areas given (A_i, D_i) . This is a simplifying assumption: Gryparis et al. (2009) show that prediction errors from spatial models are spatially correlated because nearby areas share the same monitoring station data when their surrogates are generated. Incorporating spatially correlated errors is listed as a potential extension in Section 8.

Version A: Classical error with Di-dependent mean. Following Wu et al. (2019), D_i enters through the *mean* of the surrogate, introducing systematic directional bias:

$$Z_i \mid A_i, D_i \sim \mathcal{N}(\beta_{Z1}A_i + \beta_{Z2}D_i, \sigma_Z^2). \quad (3)$$

- β_{Z1} : attenuation. $\beta_{Z1} = 1$ centres the surrogate on the truth; $\beta_{Z1} < 1$ attenuates high exposures. Wu et al. (2019) use $\beta_1 = 0.8$ as their default.
- β_{Z2} : directional bias from D_i . When $\beta_{Z2} > 0$ the surrogate overestimates true exposure in high- D_i areas (far from monitors), consistent with the MELONS finding of systematic overestimation in deprived and remote areas. When $\beta_{Z2} = 0$ this reduces to standard classical error.
- σ_Z^2 : fixed homoscedastic noise. Heterogeneity across areas is captured entirely through the mean shift $\beta_{Z2}D_i$.

Wu et al. (2019) write this in the regression calibration direction as $E(X \mid W, D) = \gamma_0 + \gamma_1 W + \gamma_2^\top D$.

Version B: Classical error with Di-dependent variance. Alternatively, D_i enters through the variance of the surrogate, relaxing the homoscedasticity assumption of Josey et al. (2023) Assumption 2:

$$Z_i \mid A_i, D_i \sim \mathcal{N}(A_i, \sigma_Z^2(D_i)), \quad \sigma_Z^2(D_i) = \exp(\eta_0 + \eta_1 D_i). \quad (4)$$

- The surrogate is unbiased on average ($E[Z_i \mid A_i] = A_i$) but noisier in high- D_i areas
- $\eta_1 = 0$ recovers Josey’s homoscedastic Assumption 2 exactly
- $\exp(\eta_0 + \eta_1 D_i)$ ensures positivity and implies a log-linear (proportional) relationship: a unit increase in D_i multiplies the error variance by $\exp(\eta_1)$, natural for distance-based quantities (Gryparis et al., 2009)
- Unlike Version A, there is no attenuation or directional bias; the error is purely a noise term whose magnitude depends on D_i

The two versions make different assumptions about how D_i influences the measurement error: Version A captures directional bias (the surrogate is systematically wrong in a direction that depends on D_i) while Version B captures variance heterogeneity (the surrogate is unbiased but less reliable in high- D_i areas). Both are scientifically motivated and are presented to the supervisor for a decision on which best reflects the CanOSSEM error structure.

4.6 Outcome Model

The area-level count outcome follows a Poisson model with a log-linear predictor and population offset, matching Josey et al. (2023):

$$Y_i \mid A_i, \mathbf{X}_i, N_i \sim \text{Poisson}(N_i \exp(\eta_i)), \quad (5)$$

$$\eta_i = -3 + f(A_i) - 0.5X_{i1} - 0.25X_{i2} + 0.25X_{i3} + 0.5X_{i4}. \quad (6)$$

- **Population offset** $\log(N_i)$: N_i scales the expected count so that $\exp(\eta_i)$ is the event rate per person. A larger area will have more events simply because it has more people; the offset removes this confound. In R this enters as an additive term $\log(N_i)$ in the linear predictor or via the `offset` argument.
- **Intercept** -3 : $\exp(-3) \approx 5\%$ baseline event rate, representing baseline depression incidence in the CLSA population (Raina et al., 2019)
- **Confounder coefficients** $\pm(0.5, 0.25)$: Follow Josey et al. (2023)
- **ERF** $f(A_i)$: Exposure-response function specified in sensitivity scenario S4; three functional forms are evaluated
- **No spatial random effect**: Consistent with thesis proposal Section 4.3; spatial correlation in outcomes is induced indirectly through the spatially correlated A_i surface

Z_i does not appear in equation (5), satisfying the non-differential assumption by construction.

4.7 Causal Estimand

The target estimand is the average potential outcome $\mu(a) = \mathbb{E}[Y_i(a)/N_i]$ (average event rate under intervention $A_i = a$ for all areas) and the causal contrast $\Delta(a_1, a_0) = \mu(a_1) - \mu(a_0)$.

- The true $\mu(a)$ is computed by Monte Carlo over $N_{\text{ref}} = 100,000$ fixed covariate vectors drawn once, held constant across all replicates
- Evaluation grid: 201 equally spaced values covering the bulk of the A_i distribution, following Josey et al. (2023)
- A pre-specified pairwise contrast (e.g. $a_1 = 15$ vs $a_0 = 5 \mu\text{g}/\text{m}^3$) is also reported as a single summary metric

Table 1: Simulation parameters. Version A and Version B refer to the two error model parameterisations in Section 4.

Parameter	Notation	Base / Range	Justification
<i>Design</i>			
Number of areas	I	200	Josey uses 400–800; reduced for BART cost
Replicates	R	500	Josey et al. (2023)
ERF evaluation points		201	Josey et al. (2023)
<i>Covariates</i>			
Area confounders	X_{ik}	$\mathcal{N}(0, 1)$	Josey et al. (2023)
Population size	$\log(N_i)$	$\mathcal{N}(8.5, 1.44)$	Statistics Canada 2021 Census
Calibration covariate	D_i	$\text{Gamma}(2, 1)$	Monitor distance proxy; Gryparis et al. (2009)
<i>True exposure</i>			
Mean	$\mu_{A,i}$	$7 + 1.5\mathbf{X}_i$	Ontario PM _{2.5} mean $\approx 7 \mu\text{g}/\text{m}^3$
SD	σ_A	3	Realistic spatial variation
Spatial correlation	ρ_A	$\{0, 0.5, 0.9\}$	0 = Josey; 0.9 = strong spatial smoothing
Spatial variance	σ_ϕ^2	0.3	Design choice
<i>Error model Version A (Classical Error)</i>			
Attenuation	β_{Z1}	$\{0.6, 0.8, 1.0\}$	1.0 = no attenuation; 0.8 = Wu et al. (2019)
Directional bias	β_{Z2}	$\{-1, 0, 1, 2\}$	0 = no bias; > 0 = overestimation (MELONS)
Noise variance	σ_Z^2	$\{1, 4, 9\}$	Low/moderate/high noise
<i>Error model Version B (variance heteroscedasticity)</i>			
Baseline variance	η_0	$\{-1, 0, 1\}$	Error at $D_i = 0$
Di-slope	η_1	$\{0, 0.5, 1.0, 1.5, 2.0\}$	$\eta_1 = 0$ recovers Josey Assumption 2
<i>Outcome</i>			
Intercept		-3	$\approx 5\%$ baseline event rate (CLSA)
ERF	$f(A_i)$	Linear/Quadratic/Non-linear	See S4
Confounder coefs	γ	$(-0.5, -0.25, 0.25, 0.5)$	Josey et al. (2023)
<i>Sensitivity scenarios</i>			
D_i distribution		$\text{Gamma}(2, 0.5); \text{Exp}(1)$	Wider spread; Southern Ontario
N_i distribution		$\text{Uniform}(10, 1000)$	Josey comparability
Exposure distribution		Log-normal	Extreme wildfire events
Number of areas	I	$\{100, 200, 400\}$	Scaling

5 Simulation Parameter Grid

6 Sensitivity Scenarios

S1: D_i distribution. Gamma(2, 0.5): mean 4, more heterogeneity between areas. **Exponential(1):** monotone decreasing, consistent with Southern Ontario station concentration. **Empirical:** actual FSA-to-NAPS distances from Statistics Canada shapefiles.

S2: Error model alternatives. Version A vs Version B: The two parameterisations in Section 4 are compared directly to assess whether directional bias (Version A) or variance heteroscedasticity (Version B) better characterises CanOSSEM error structure.

S3: Spatial correlation. Scenario B ($\rho_A = 0$): spatially independent exposures. Directly comparable to Josey et al. (2023) which sets $\phi_i = 0$ for all areas. **Scenario A** ($\rho_A = 0.9$): high spatial smoothing, reflecting a sustained regional wildfire event where nearby FSAs share similar smoke concentrations.

S4: Exposure-response functional forms. Three forms spanning simple to complex, crossed with spatial scenarios (S3) to give six main simulation cells.

Linear: $f(A_i) = 0.10(A_i - 7)$. Monotone increasing with no curvature. Coefficient 0.10 rescaled from Josey’s 0.25 for the Ontario exposure scale. The workhorse of air pollution epidemiology (Pope et al., 2002).

Quadratic: $f(A_i) = 0.15(A_i - 7) - 0.008(A_i - 7)^2$. Concave shape that rises steeply then flattens at high concentrations. Biologically motivated by potential saturation of health pathways at high $\text{PM}_{2.5}$.

Nonlinear (Josey cosine form): $f(A_i) = 0.10(A_i - 7) - 0.75 \cos(\pi(A_i - 5)/8) - 0.25(A_i - 7)X_{i1}$. S-shaped curve with an exposure-covariate interaction. The interaction means the ERF shape differs across areas depending on X_{i1} , which BART can capture but parametric GLMs cannot. Primary misspecification stress test.

S5: Log-normal exposure. $\log(A_i) \sim \mathcal{N}(1.68, 0.54)$ matching Ontario moments: median 5.4, mean 7, SD 5.9. Captures extreme wildfire events (99th percentile $\approx 26 \mu\text{g}/\text{m}^3$) that the Normal cannot represent. ERF evaluation grid adapts to the 5th–95th percentile.

S6: Number of areas. $I \in \{100, 200, 400\}$. All must be perfect squares for the synthetic grid.

7 Performance Metrics

For each scenario and method (naive, oracle, proposed, regression calibration):

- **Bias:** $\hat{\mu}(a) - \mu(a)$ averaged over evaluation points and replicates
- **RMSE:** $\sqrt{R^{-1} \sum_r (\hat{\mu}_r(a) - \mu(a))^2}$
- **Coverage:** fraction of 95% credible intervals containing $\mu(a)$
- **Interval width:** mean width of 95% credible intervals
- **Causal contrast bias:** $\hat{\Delta}(a_1, a_0) - \Delta(a_1, a_0)$ for a pre-specified pair

Pointwise results at $a = 11 \mu\text{g}/\text{m}^3$ are also reported following Josey et al. (2023) Table 1.

8 Potential Extensions

- **Spatially correlated measurement errors:** The current model assumes conditional independence of measurement errors $\varepsilon_i = Z_i - \mathbb{E}[Z_i | A_i, D_i]$ across areas which is a simplifying assumption that Gryparis et al. (2009) show is violated in practice: prediction errors from spatial models are correlated across nearby areas because those predictions are generated using the same monitoring station data. A natural extension is to give the error term its own spatial covariance structure, either by making D_i spatially structured via a second CAR prior (so that nearby areas have correlated error variances), or by adding an explicit spatially correlated residual $\psi_i \sim \text{CAR}(\rho_\varepsilon, \sigma_\varepsilon^2)$ to the error model.
- **Individual-level binary outcome:** Replacing the area-level Poisson count with individual-level Bernoulli outcomes $Y_{ij} \sim \text{Bernoulli}(p_{ij})$ and individual covariates L_{ij} , matching the CLSA data structure. The Bayesian bootstrap would draw Dirichlet weights over all $N = \sum_i n_i$ individuals rather than over areas.
- **Spatial random effect on outcome (ν_i):** A second CAR prior capturing residual spatial clustering in health outcomes (Englert et al., 2025). Left out of the base model per thesis proposal Section 4.3.
- **Negative binomial outcome:** The Poisson model assumes $\text{Var}(Y_i) = \mathbb{E}[Y_i]$, which is often violated in practice when area-level counts are overdispersed due to unmeasured heterogeneity across FSAs. A negative binomial outcome relaxes this by introducing a dispersion parameter $\xi > 0$:

$$Y_i | A_i, \mathbf{X}_i, N_i \sim \text{NegBin}(\mu_i, \xi), \quad \mu_i = N_i \exp(\eta_i),$$

where $\mathbb{E}[Y_i] = \mu_i$ and $\text{Var}(Y_i) = \mu_i + \mu_i^2/\xi$. As $\xi \rightarrow \infty$ this recovers the Poisson; small ξ gives strong overdispersion. Equivalently this arises from a Poisson-Gamma hierarchy as in Englert et al. (2025): $Y_i | \theta_i \sim \text{Poisson}(\theta_i)$ where $\theta_i \sim \text{Gamma}(\xi, \mu_i/\xi)$, marginalising over θ_i to give the NegBin. The dispersion parameter $\xi = 1$ is a natural starting point and is the value used by Englert et al. (2025) in their simulation. This extension tests whether the proposed method remains well-calibrated when the outcome model is more variable than the base Poisson.

- **Zero-inflated outcome:** For rare outcomes or small populations where structural zeros arise.
- **Log-normal exposure model:** The base model specifies A_i on the original scale as a Normal with a floor at zero. An alternative motivated by the right-skewed nature of PM_{2.5} concentrations and by the suggestion that $\log(A_i)$ should be regressed on covariates is to model the log-exposure directly:

$$\log(A_i) | \mathbf{X}_i, \phi_i \sim \mathcal{N}(\gamma_0 + \gamma_1 X_{i1} + \gamma_2 X_{i2} + \gamma_3 X_{i3} + \gamma_4 X_{i4} + \phi_i, \sigma_{\log A}^2),$$

so that $A_i = \exp(\log A_i) > 0$ always. The parameters are matched to Ontario PM_{2.5} empirical moments: setting $\gamma_0 = 1.68$ and $\sigma_{\log A} = 0.73$ gives $\mathbb{E}[A_i] \approx 7 \mu\text{g}/\text{m}^3$ and $\text{SD}(A_i) \approx 5.9 \mu\text{g}/\text{m}^3$, matching the provincial monitoring data. The error model in Version A would then operate on the log scale as $\log(Z_i) | \log(A_i), D_i \sim \mathcal{N}(\beta_{Z1} \log(A_i) + \beta_{Z2} D_i, \sigma_Z^2)$, giving multiplicative error on the original scale. The causal estimand $\mu(a)$ is computed at values of a on the original scale, but the exposure-response curve is evaluated over a grid adapted to the 5th–95th percentile of the log-normal distribution rather than a fixed interval.